

1. IP: static IP vs. dynamic IP; public IP vs. private IP

The following are the commercialized advertisement of a ISP about their products on internet access services:

ADSL上網服務可概分為「固定制」與「非固定制」兩種取得IP的方式。
ADSL「固定制」指ISP直接配發數個「固定的真實IP」給客戶，
使用時只要將固定IP直接設定在電腦或是網路設備上，便可達到「開機即上網」的目的。
至於ADSL「非固定制」則透過「PPPoE」或「DHCP」等方式提供上網服務，
由於每次ISP配發的IP並不固定，故稱為「非固定制」。

- (a) What are the differences between static IP and dynamic IP?
- (b) In what situation we had better to use static IP?
- (c) In what situation using a dynamic IP is just good enough?
- (d) What are the differences between public IP and private IP?
- (e) In what situation we had better to use public IP?
- (f) In what situation using a private IP is just good enough?

2. IP address assignment

Suppose we have a subnet: 1.2.2.0/23,

- (a) How many IP addresses in this subnet, which we can assign to individual network devices?
- (b) Two IP addresses, the first (1.2.2.0/23) and the last (1.2.3.255/23), in this subnet we should reserve for special use. what is the first IP (1.2.2.0/23) for? and what is the last IP (1.2.3.255/23) for?

3. IP transmission

Refer to Figure 1.

- (a) According to figure 1, what is the gateway address for Host A (IP=1.2.3.10/24), and what is the net mask for Host A?
- (b) New suppose host A mis-config its net mask to a bigger one (255.255.254.0), under this assumption, can host A communicates with B successfully? Similarly, can host A communicates with C and G successfully?
- (c) New suppose host A mis-config its net mask to a smaller one (255.255.255.254), under this assumption, can host A communicates with B successfully? Similarly, can host A communicates with C and G successfully?

4. What are differences between LS (link state) and DV (distance vector) routing strategies?
5. What is the link state routing oscillation problem? and how to prevent it? What is count-to-infinity problem with distance vector routing strategy? And how to prevent it?
6. Give an example about how the CRC-checksum works.

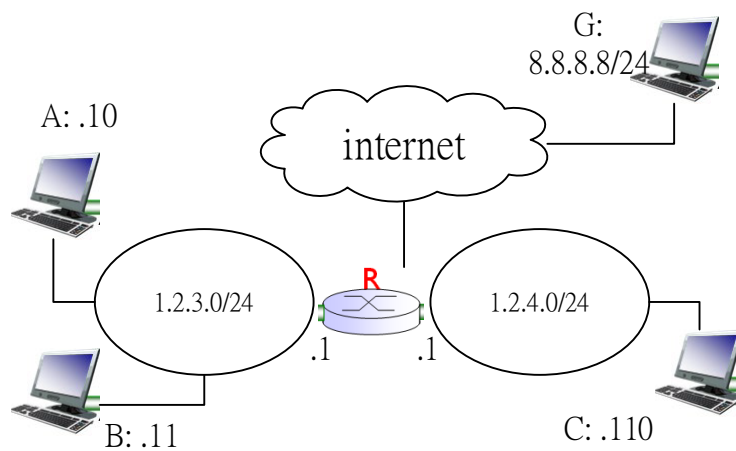


Figure 1